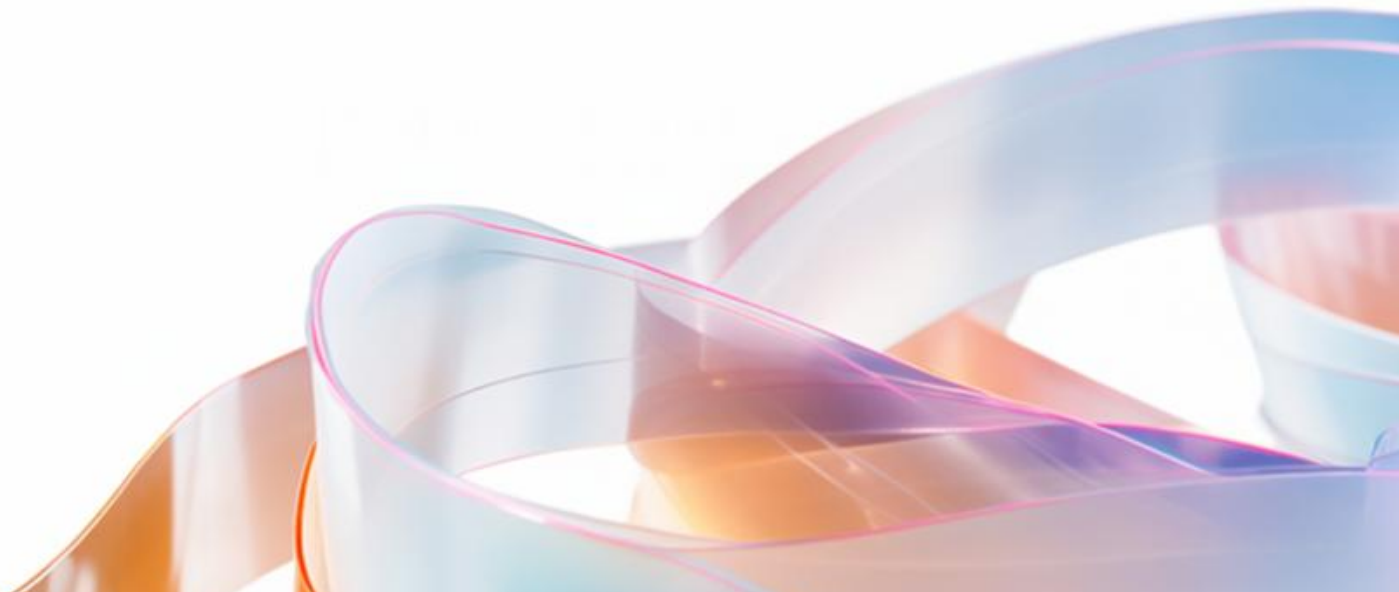


Are your semantic models ready for Copilot?

Lars Andersen
Microsoft
Fabric CAT





18th edition SQLDay Conference



11-13 May 2026, WROCŁAW + ONLINE

Platinum sponsor



Gold sponsors



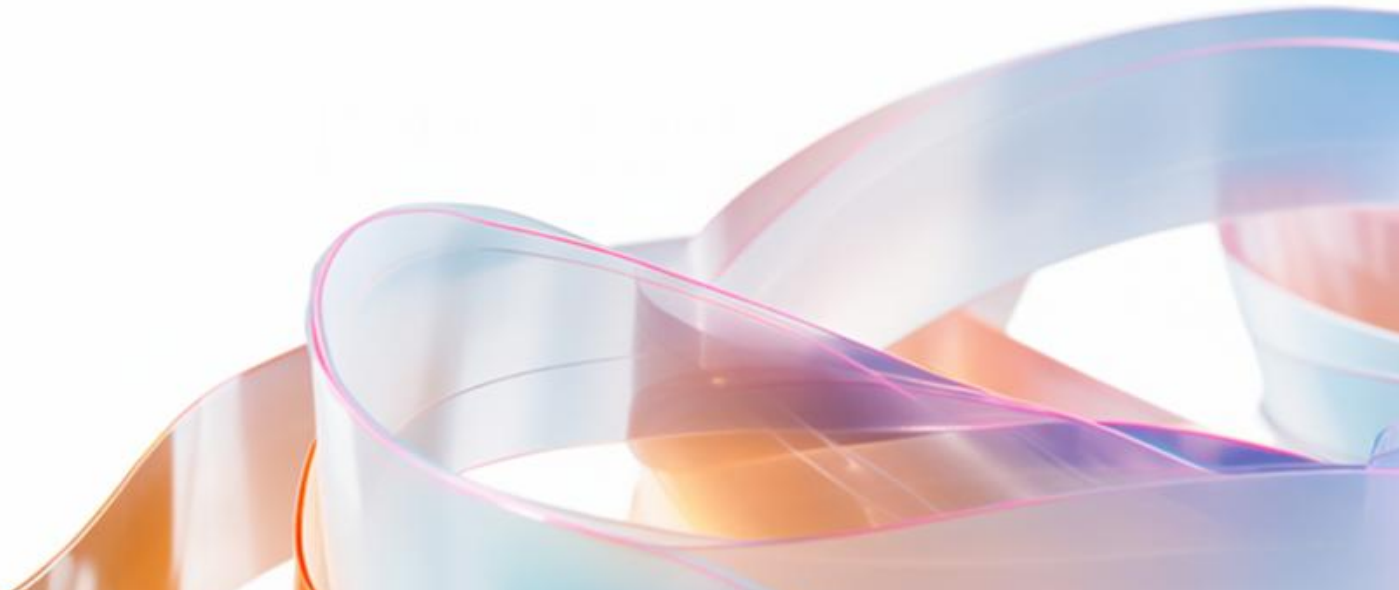
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Agenda

- Data answering in Fabric
- Enabling Copilot for Power BI
- Understand how to use Copilot
- Grounding data for Copilot
- Best practice and prepare semantic models for Copilot

Data answering in Fabric



Data Answering

Business user AI experiences in Fabric



Copilot for
Power BI

Generalist



Fabric Data
Agents

Specialist

Similarities

You may have noticed some key similarities



Copilot for Power BI



Data Agents

- ✓ Highly conversational
- ✓ Answer user questions about their data
- ✓ Human in the loop setup for best results

Different Goals

Built to help customers get data insights using AI



Copilot for Power BI

Extract **more** value & new insights from **existing data** content & context



Data Agents

Craft **new** agents to deliver a **tailored experience** on top of OneLake

Scope

What data can be reasoned over?



Standalone Copilot

- Visual layer: All Reports
- Data layer: Semantic models
 - Data agents
- *All* content the user has access to PBI



Data Agents

- **Only** data layer: Structured & unstructured, OneLake & mirrored sources, ontologies, semantic models
- Just what is added to the data agent
- Limit to 5 data sources (SDK not limited)

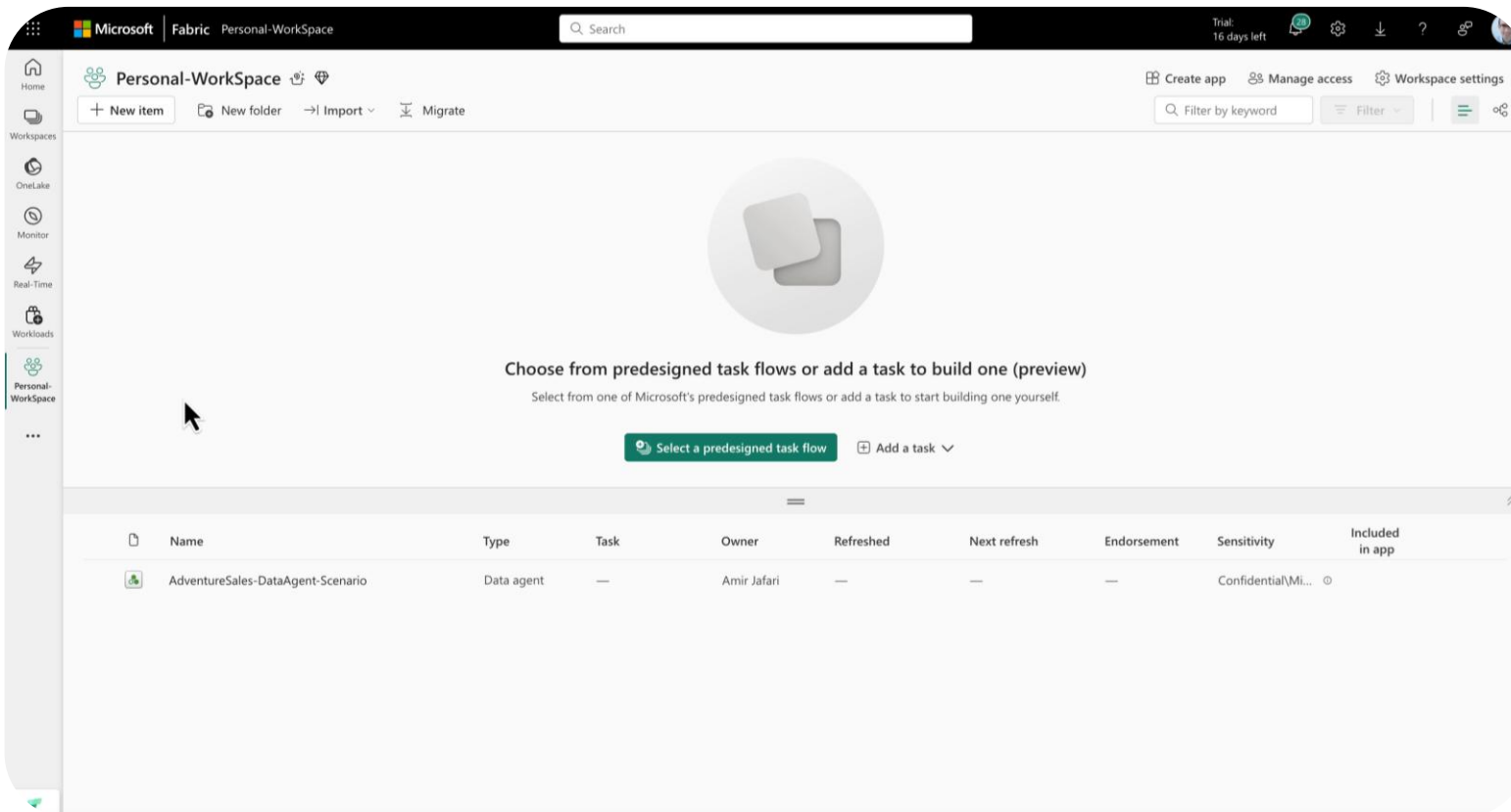
Fabric Data Agent

Enrich custom copilots with Fabric data expertise

Create a powerful data expert **tailored to reason** over and synthesize data in OneLake for insights.

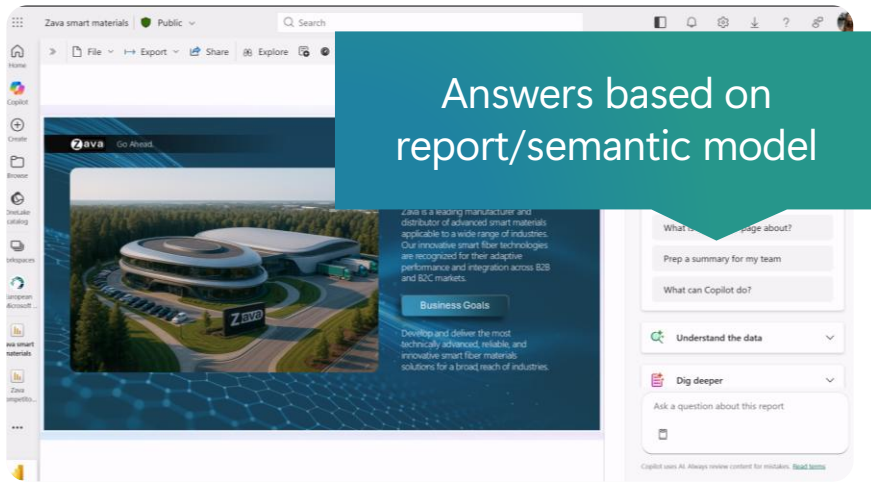
Leverage **structured & unstructured data sources** to provide holistic answers.

Security permissions, including row and column level security, **are respected**.

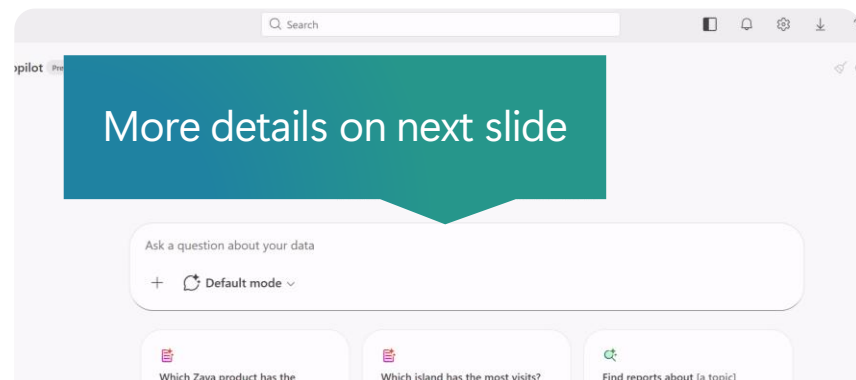


Copilot for Power BI

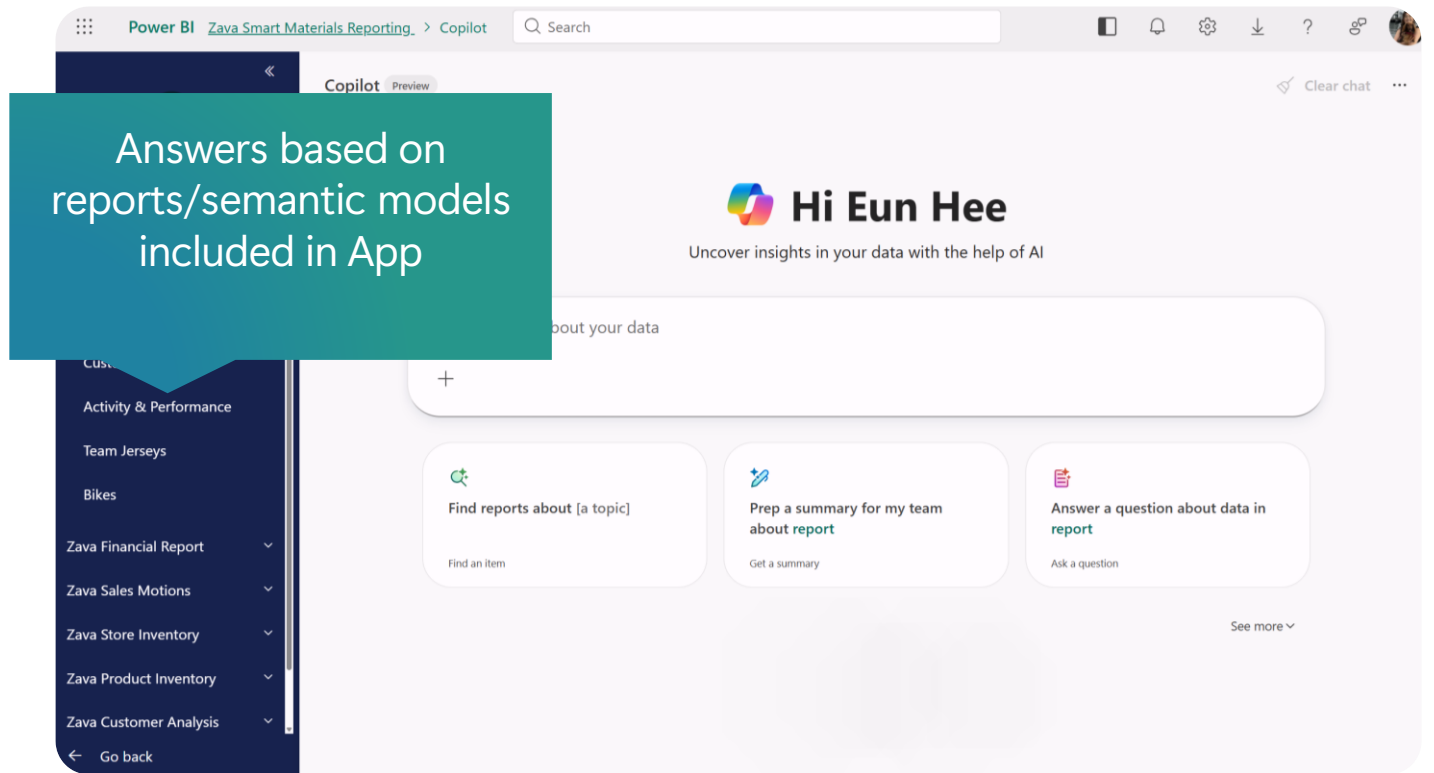
Report Pane



Standalone



Workspace Apps



Power BI standalone Copilot

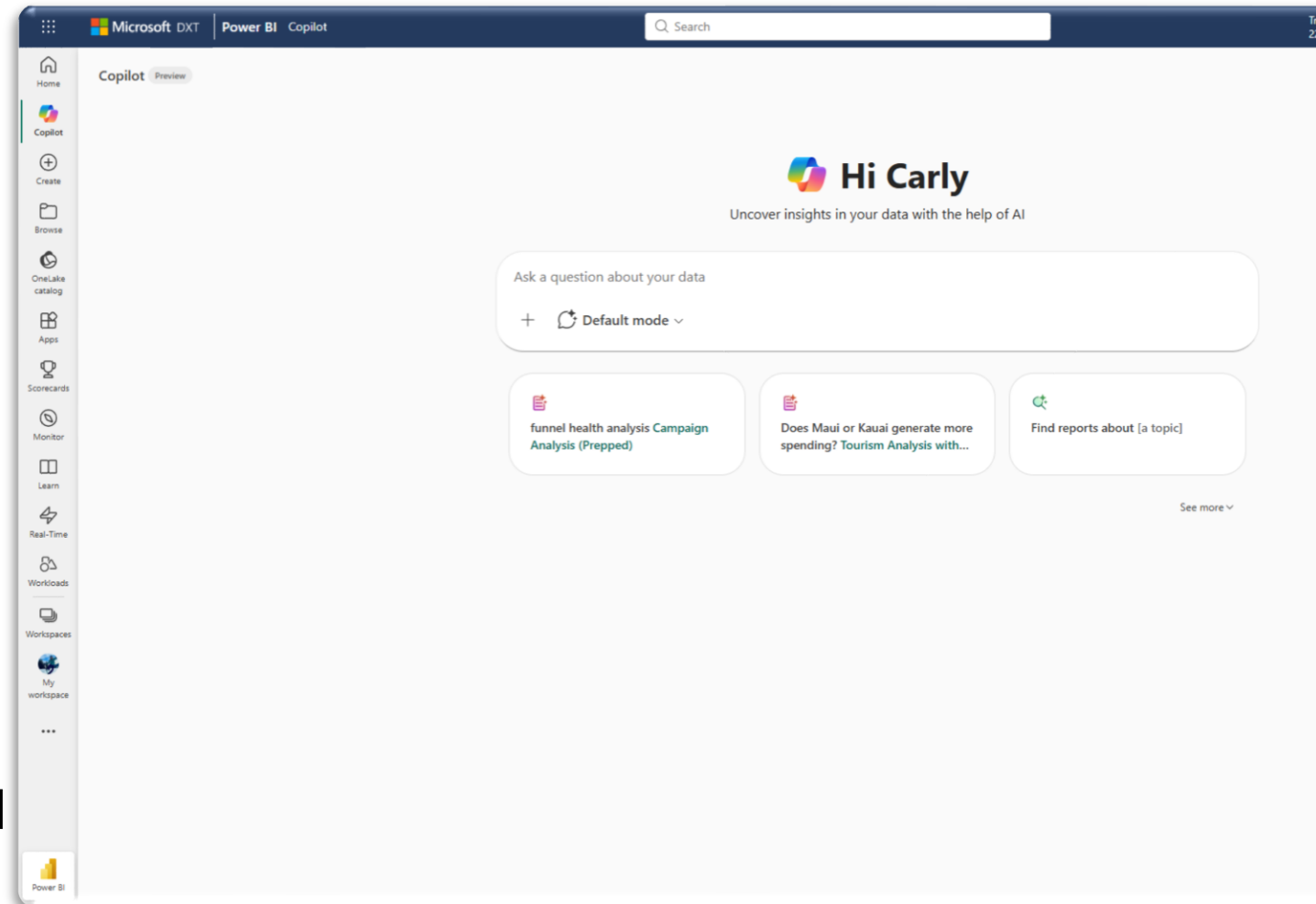
A full-screen Copilot experience that enables chat with your data

Interactive visuals results and narrative summaries to **answer data questions**.

Finds the right data source across all semantic models, reports, apps, and data agents.

Visuals can be explored or saved to **build with Copilot's help**.

Security permissions, including row and column level security, **are respected**.



Side-by-side comparison

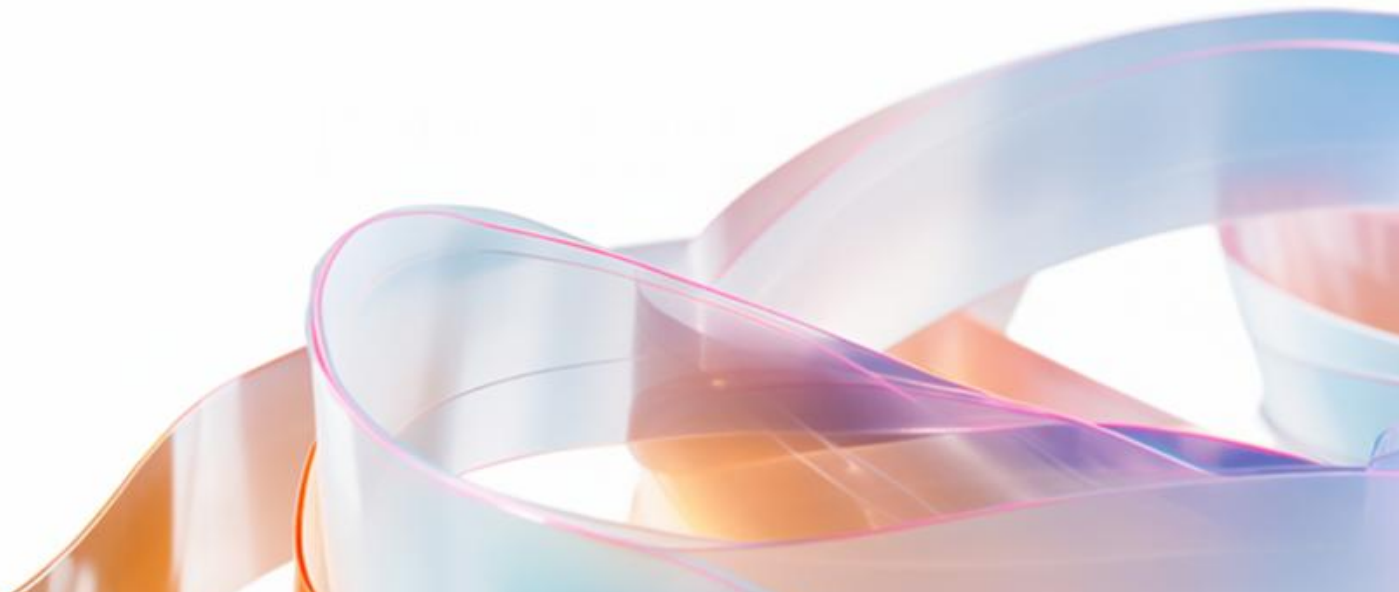


Scenario	Power BI Standalone Copilot	Fabric Data Agent
Explore a single dataset	✓ Best choice	⚠ Overkill
Build new / explore visuals	✓	✗
Utilize business metrics and logic	✓ Strong	⚠ Depends on sources
Cross-data-source questions	✗ Limited	✓ Strong
Decision support / recommendations	⚠ Light	✓ Strong
Automation / workflows	✗	✓
Unstructured + structured data	✗	✓
Embedded in apps/solutions	✗	✓

Scope for the rest of this session



Enabling Copilot for Power BI



Tenant settings

- Enabling of Copilot in Power BI is documented here [Enable Fabric Copilot for Power BI](#)
- As of today, these tenant settings exist:

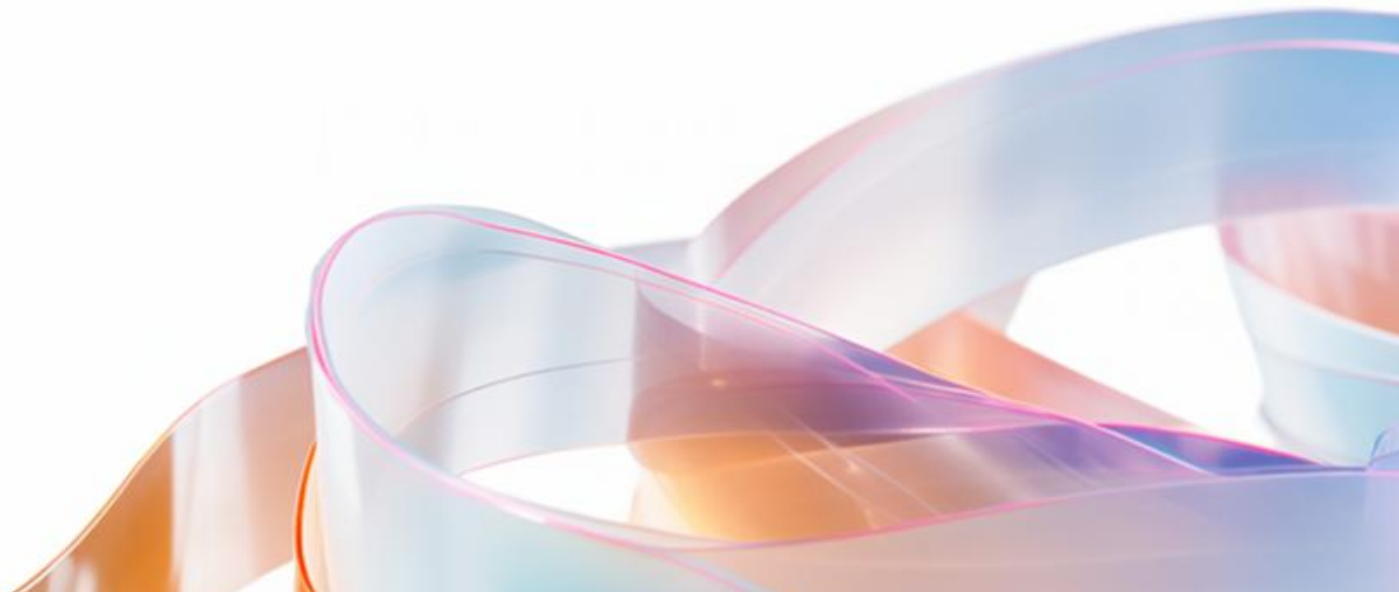
Copilot and Azure OpenAI Service

- Users can use Copilot and other features powered by Azure OpenAI
Enabled for the entire organization
- Users can access a standalone, cross-item Power BI Copilot experience (preview)
Enabled for the entire organization
- Data sent to Azure OpenAI can be processed outside your capacity's geographic region, compliance boundary, or national cloud instance
Enabled for the entire organization
- Capacities can be designated as Fabric Copilot capacities
Enabled for the entire organization
- Data sent to Azure OpenAI can be stored outside your capacity's geographic region, compliance boundary, or national cloud instance
Enabled for the entire organization
- Only show approved items in the standalone Copilot in Power BI experience (preview)
Disabled for the entire organization

Tenant settings

- *Users can use Copilot and other features powered by Azure OpenAI: **Must be enabled to use Copilot***
- *Users can access a standalone, cross-item Power BI Copilot experience (preview): **Optional***
- *Data sent to Azure OpenAI can be processed outside your capacity's geographic region, compliance boundary, or national cloud instance: **Required if Azure OpenAI isn't available in capacity region***
- *Capacities can be designated as Fabric Copilot capacities: **Optional***
- *Data sent to Azure OpenAI can be stored outside your capacity's geographic region, compliance boundary, or national cloud instance: **Optional, applicable for Copilot in Notebooks and Data agent feature***
- *Only show approved items in the standalone Copilot in Power BI, and in the Power BI Agent experiences (preview): **Optional***

Understand how to use Copilot



Understand how to use Copilot

Enable Q&A

Allows Copilot to answer questions over your data

Understand limitations

Report > Model

Supported v.
unsupported query
types

Perfecting questions

Ideal sentence
structure

Understand how to use Copilot

Enable Q&A

Allows Copilot to answer questions over your data



Copilot

AI-generated content may be incorrect

To help me respond to requests like this, turn on Q&A for this semantic model.

Understand how to use Copilot

Enable Q&A

Allows Copilot to answer questions over your data

Q&A

① If Q&A is turned on, we'll create an index of your data in the service.[Learn more](#)

☒ Turn on Q&A to ask natural language questions about your data

☐ Share your synonyms with everyone in your org

Apply

Discard

Understand how to use Copilot

Copilot limitations

Report response

Report visuals are prioritized over the model information

The data shows the number of leads generated by each ad type:

- **Site:** 677.44K leads ¹
- **Blogs:** 666.11K leads ¹
- **Magazine:** 657.55K leads ¹
- **TV:** 599.57K leads ¹

Therefore, the ad type that has the most leads is **Site** with 677.44K leads.

Is this response helpful?

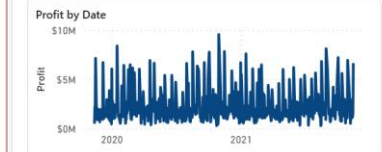


which ad type has most leads?

Model response

Show me profits over time

Showing date and profit.



Explore this visual

How Copilot arrived at this

You asked: "Show me profits over time."

Data used

Date

Profit

Filters applied 1

None

Is this response helpful?



Understand how to use Copilot

Copilot limitations



Columns and fields in the report or data model



Explanations, summaries, or “show me” questions

“What were the top 5 selling products in North America last month?”

where product, region, and date are already columns in the data, and Copilot helps with top N filtering to produce the visual.

“Tell me the average price of gasoline per gallon over the last 30 days.”

where price per gallon is a field already contained in the model, and Copilot helps by taking the average and filtering by relative date to produce the visual.

Understand how to use Copilot

Copilot limitations

x

Why questions

x

Diagnostic analysis like
root cause or
forecasting

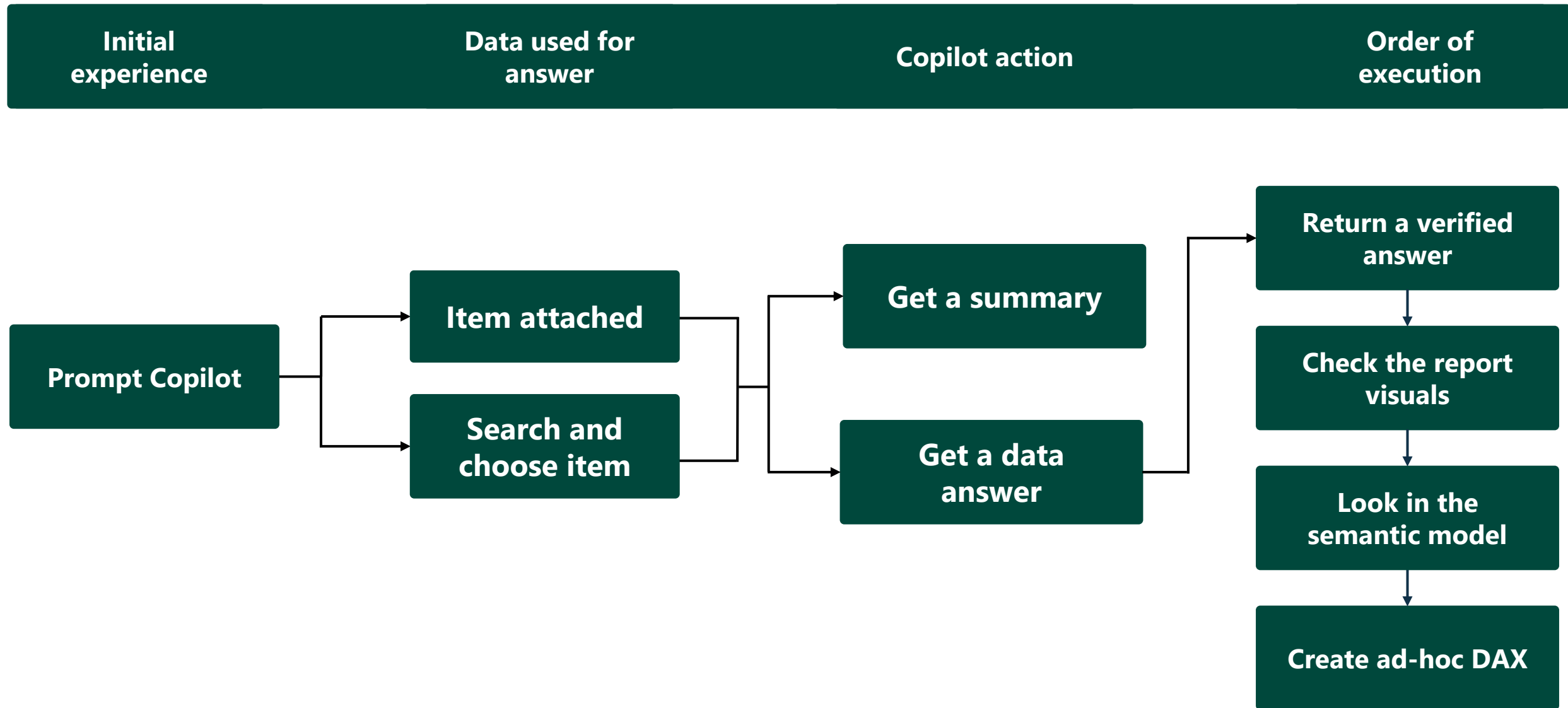
**"How many books do you
think we will sell next year?"**

This question asks for forecasting,
which isn't currently supported

**"Why do our sales go down
every July?"**

This question involves generating
deeper insights beyond the provided
data.

Behind the scenes (simplified)



Understand how to use Copilot

Perfecting questions

Ideal sentence structure

Average ticket sales by movie genre

... grouped by region

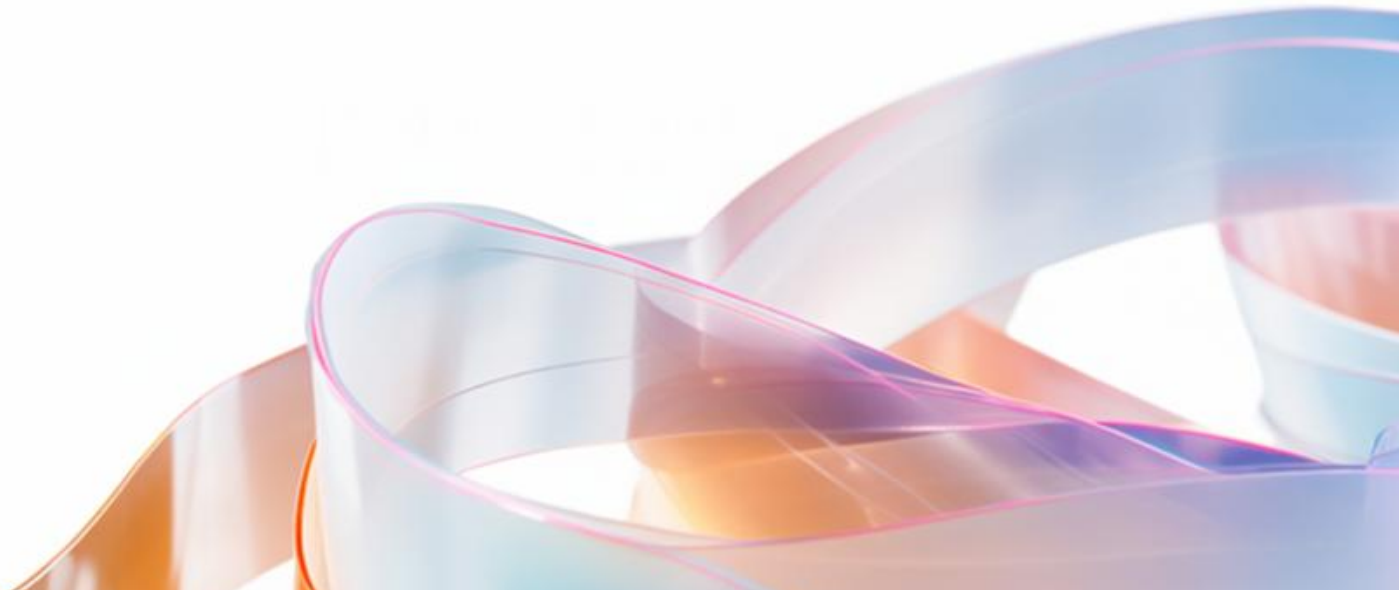
... released in 2024

Top 5 movie titles by total ticket sales

... as a column chart

1. preferred sort order
2. aggregation on value/measure by dimension field
3. legend
4. filters
5. visual type

Grounding data for Copilot



Configuring and grounding data for Power BI Copilot:

Prep data for AI

Verified Answers

Ensure **quality** and **consistency** of responses for the most critical business topics.

Schema Selection

Limit the scope of data that Copilot can answer questions about

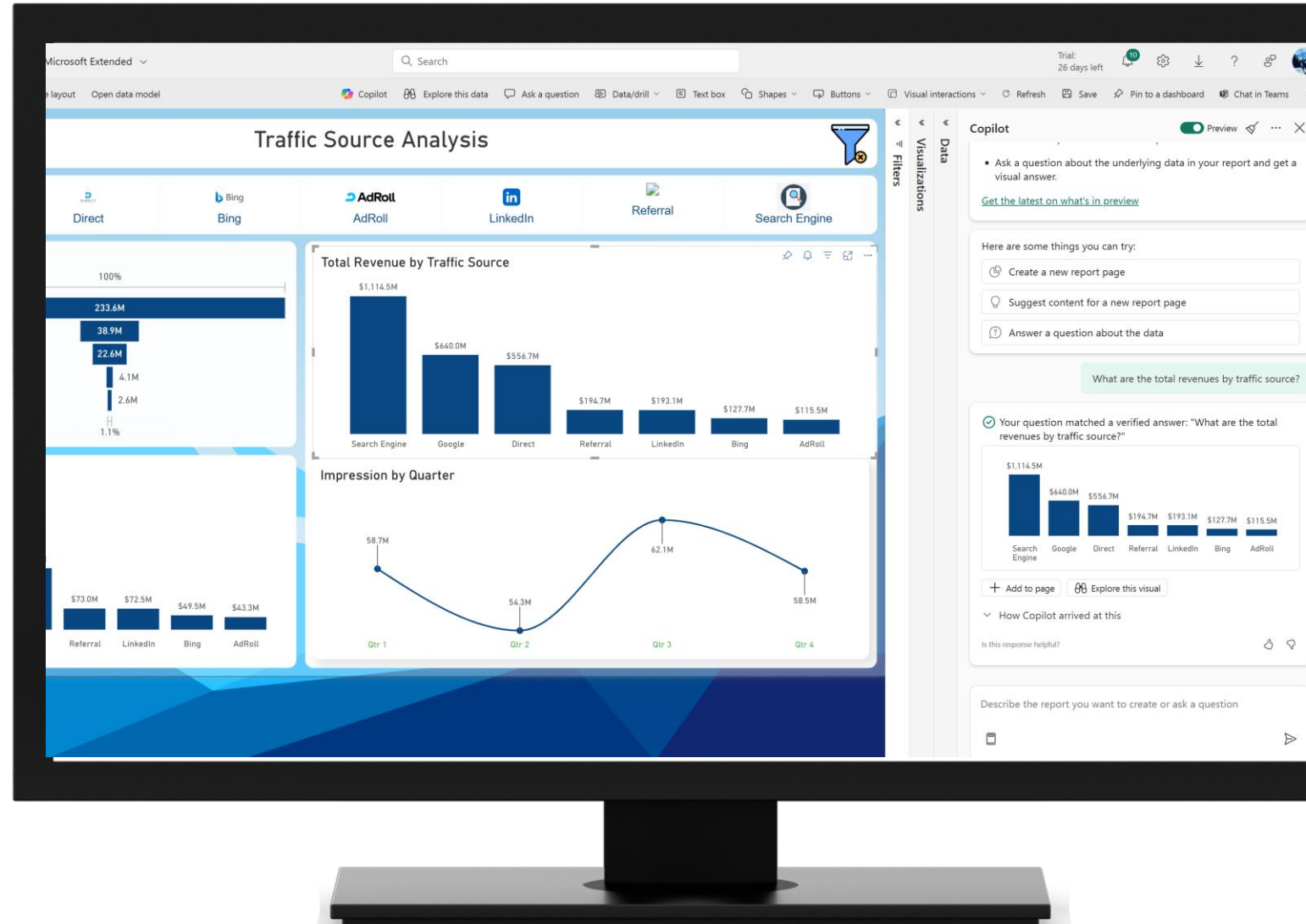
Custom Instructions

Provide business context and help Copilot guide how to talk about your data

Power BI Copilot grounding: Verified Answers

Verified Answers

Ensure **quality** and **consistency** of responses for the most critical business topics.

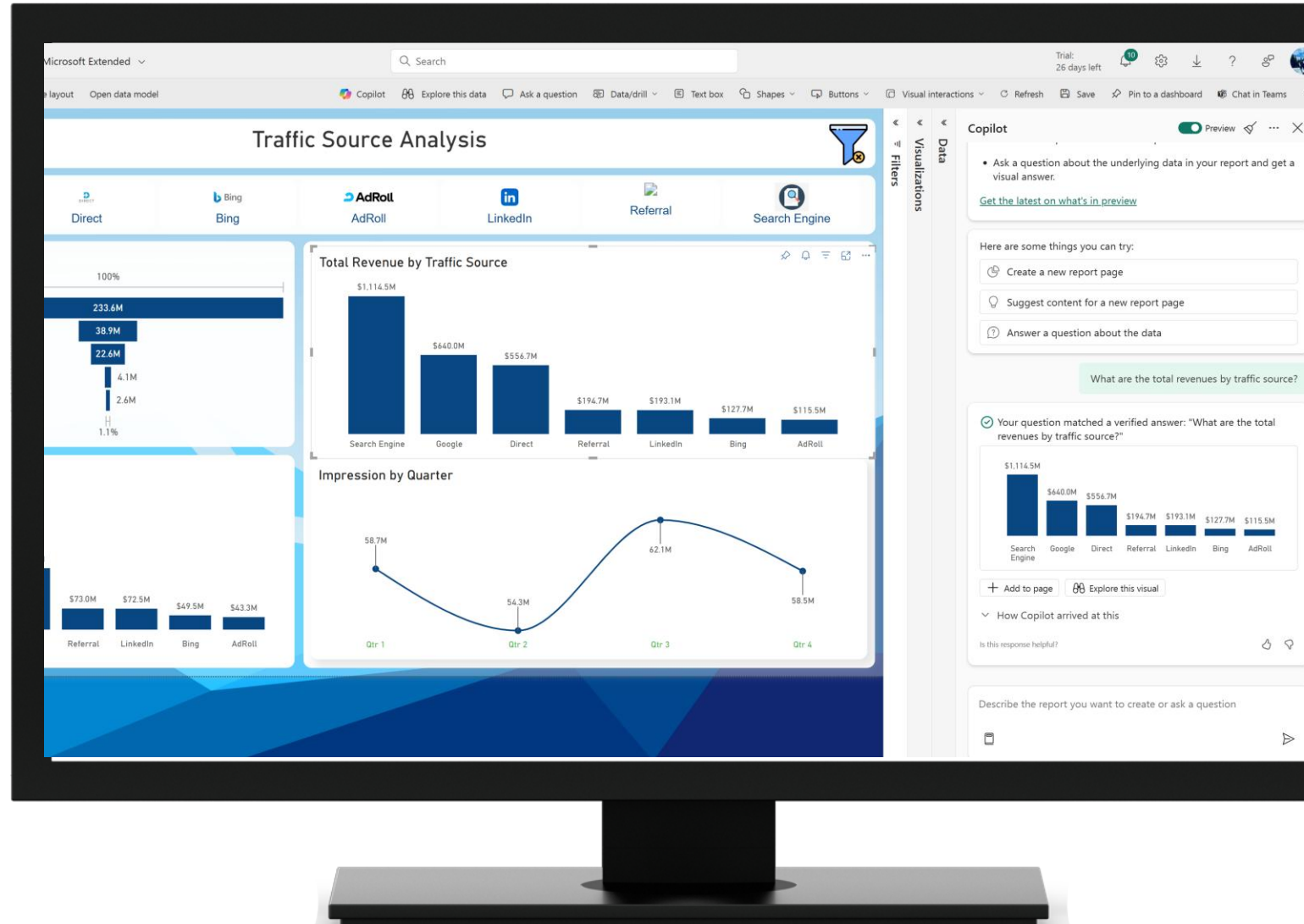


Benefits

Verified Answers

Benefits

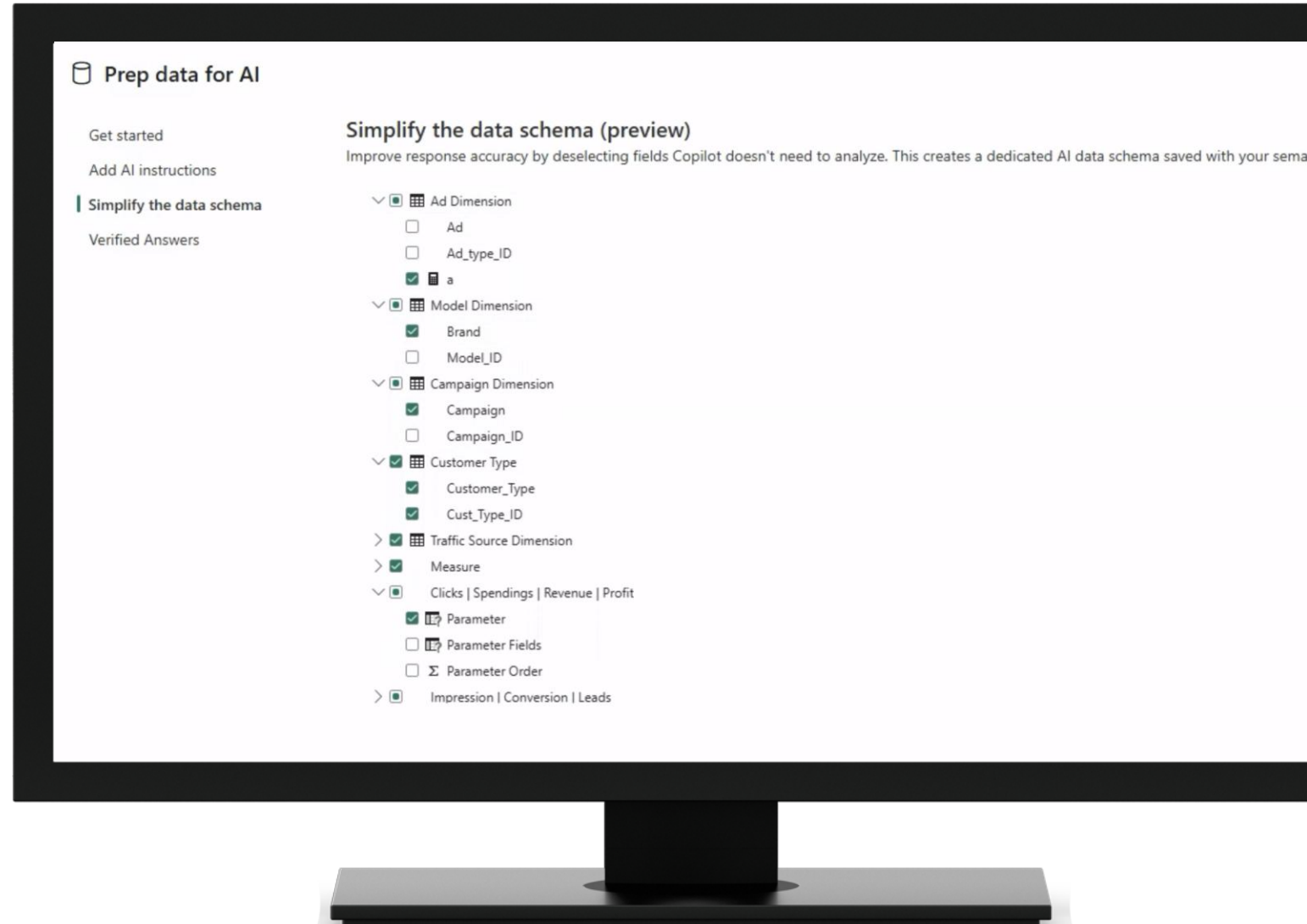
- Accuracy + consistency
- Capture business context
- Help Copilot get smarter over time
- Reuse accurate responses across models



Power BI Copilot Grounding: Schema Selection

Schema Selection

Schema selection allows users to prioritize specific data schemas for more relevant and focused insights.



Benefits

Schema Selection

Benefits

- Column and **measure** level control
- Enhance relevance of specific measures
- Ensure Copilot model responses are friendly and pointing to the right data

Prep data for AI

Get started

Add AI instructions

Simplify the data schema

Verified Answers

Simplify the data schema (preview)

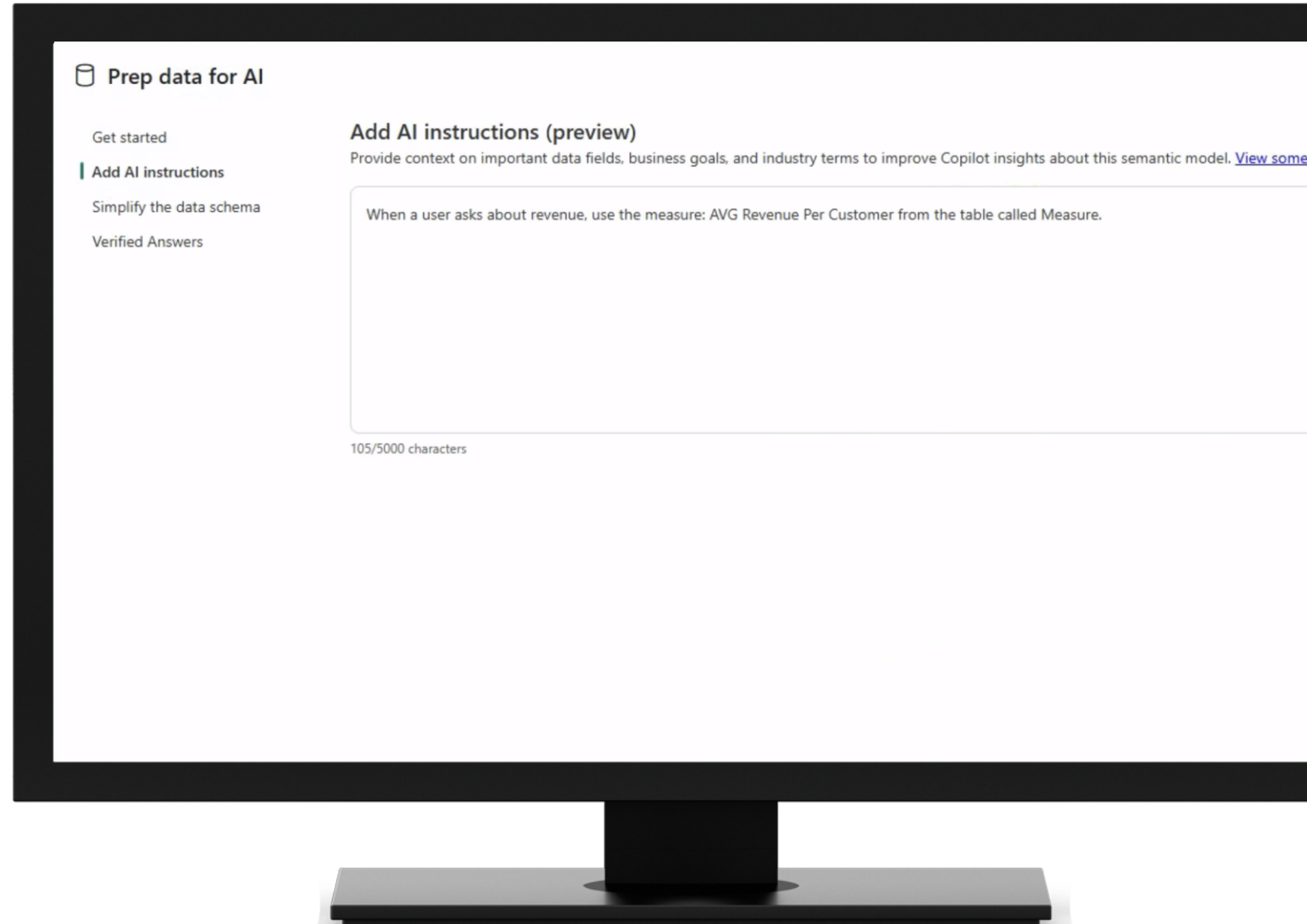
Improve response accuracy by deselecting fields Copilot doesn't need to analyze. This creates a dedicated AI data schema saved with your semantic model.

- ✓ ☒ Ad Dimension
 - ☐ Ad
 - ☐ Ad_type_ID
 - ☒ a
- ✓ ☒ Model Dimension
 - ☒ Brand
 - ☐ Model_ID
- ✓ ☒ Campaign Dimension
 - ☒ Campaign
 - ☐ Campaign_ID
- ✓ ☒ Customer Type
 - ☒ Customer_Type
 - ☒ Cust_Type_ID
- > ☒ Traffic Source Dimension
- > ☒ Measure
- ✓ ☒ Clicks | Spendings | Revenue | Profit
 - ☒ Parameter
 - ☐ Parameter Fields
 - ☐ Parameter Order
- > ☒ Impression | Conversion | Leads

Power BI Copilot Grounding: Custom Instructions

Custom instructions

Customize Copilot with instructions for how to best respond to your data.

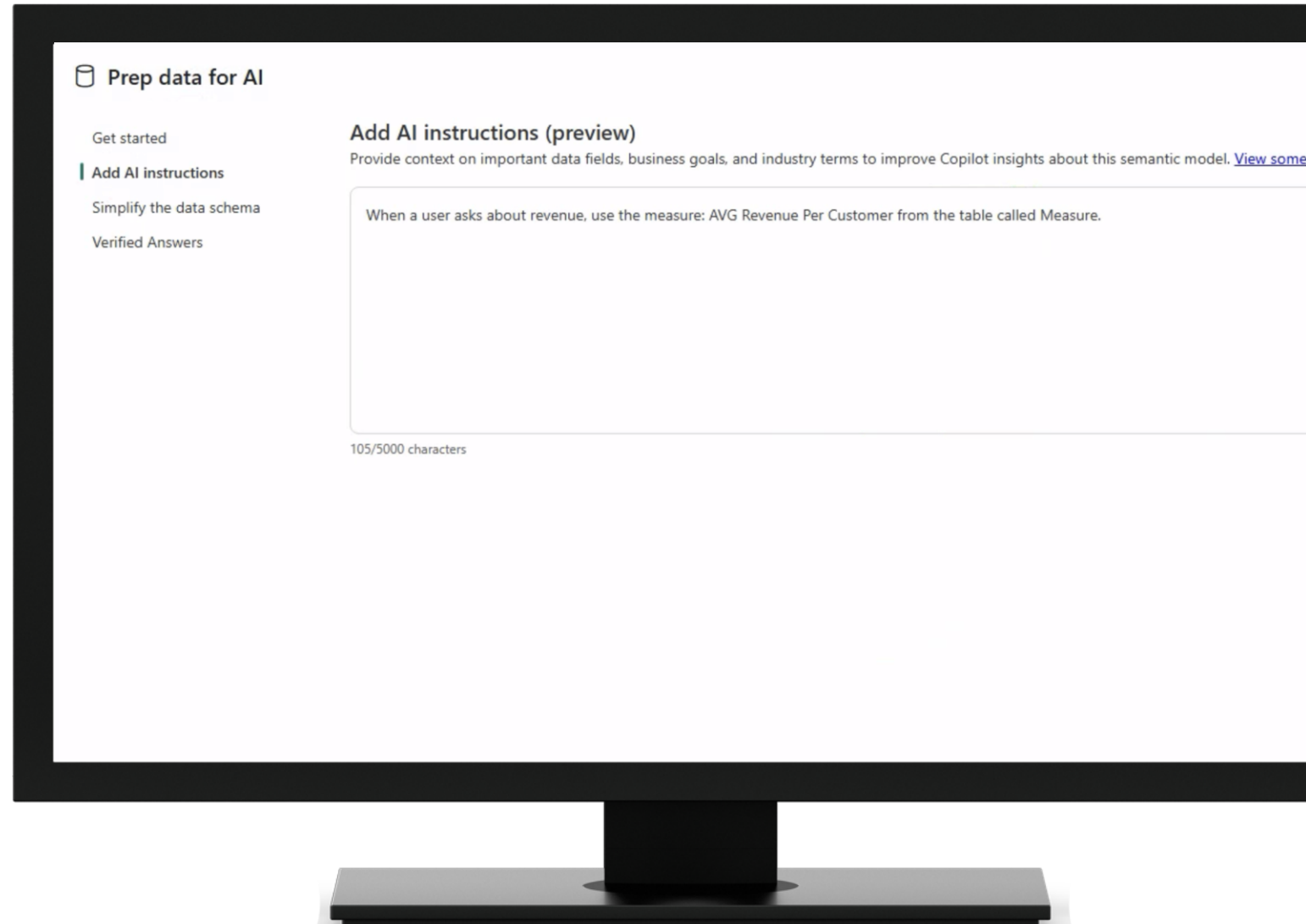


Benefits

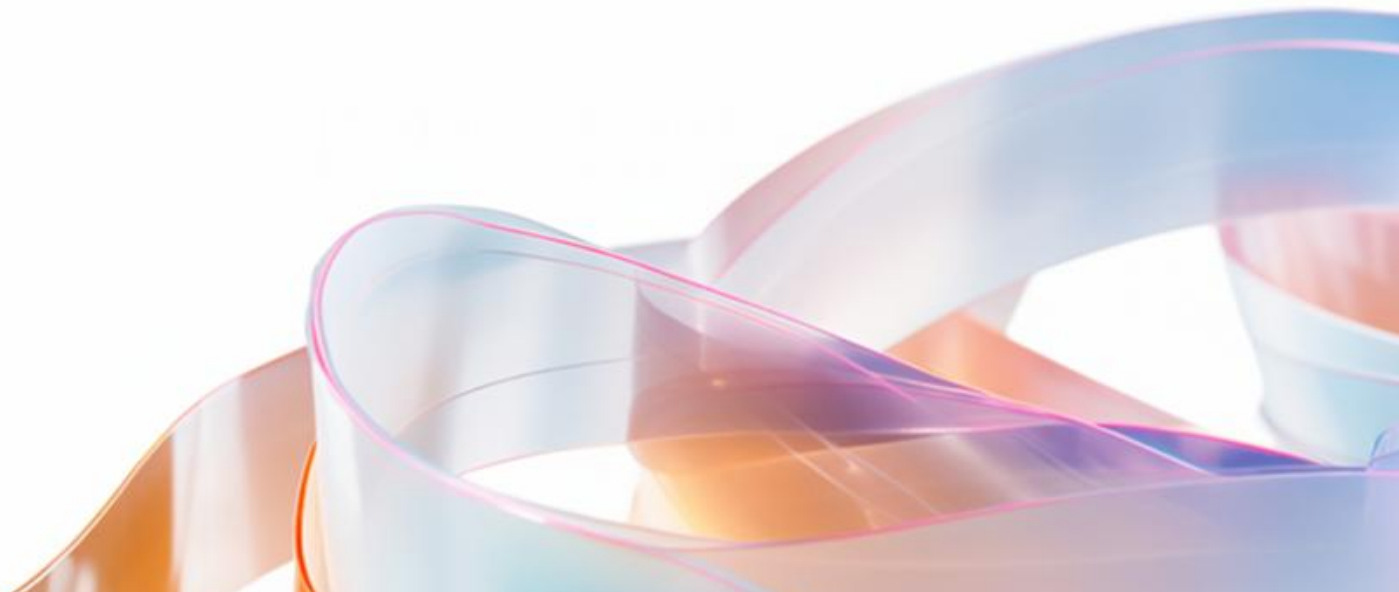
Custom Instructions

Benefits

- Personalized Experience: Tailor Copilot's responses with instructions for how to best respond to your data.
- Enhanced Efficiency: Streamline workflows by providing specific instructions that align with user goals.
- Improved Accuracy: Ensure Copilot understands and executes tasks correctly by offering clear guidance.



Best practice and prepare semantic models for Copilot



Best practices – how to get the most of Copilot

- Try and be verbose as it will better understand your request (examples: be specific on field names)
- Steer Copilot, add clarifications to guide Copilot to the correct answer
- Ask follow up questions
- Its an LLM, it can still use ungrounded reasoning
- If something is offensive, it will then return an error as Copilot
- Use thumbs up / thumbs down for feedback to Microsoft

Modeling best practices

Copilot's performance is tightly linked to the quality of your semantic model.

Key practices include

- **Clear human-readable naming conventions:** Avoid ambiguous or overly similar field names. Consistency improves Copilot's ability to interpret prompts accurately.
- **Defined relationships:** Ensure relationships are well-structured and logical (star schema!!). Poorly defined joins or missing keys degrade Copilot's output.
- **Create core measures:** Build essential DAX measures (e.g., Total Sales, Profit Margin, Customer Count) so Copilot has strong building blocks to answer questions.
- **Metadata enrichment:** Add descriptions to model objects, fields, etc. This helps Copilot understand context and improves prompt responses.

Prep data for AI – Best Practices

- **Simplify the data schema** – 7 tips
- **Verified answers** – 12 tips
- **AI instructions** – 7 tips

Simplify the data schema

Helps Copilot to focus on the right data by only exposing what Copilot needs to see

1. **Visibility level** - Consumers of your semantic model can't see the AI data schema set on the model.
2. **Scope of capability** - The AI data schema isn't considered for Copilot capabilities that don't use the model schema. For example, when Copilot is giving a summary of the report or answering a question directly from the report page.
3. **Hidden state** - All fields available in the semantic model appear in the AI data schema selection pane regardless of visibility set on the semantic model.

Simplify the data schema (preview)

Improve response accuracy by deselecting fields Copilot

🔍 Filter by keyword

▼ ☒  FIAD - final for standalone

> ☐  Customer

> ☒  Date

> ☒  Geo

> ☐  People

> ☒  PO

> ☒  Product

> ☒  ProductDetails

> ☒  Reseller

> ☒  Sales

> ☒  Supplier

Simplify the data schema (continued)

Helps Copilot to focus on the right data by only exposing what Copilot needs to see

4. **Hierarchies always respected** - If a hierarchy is included in the AI data schema, but a column of the hierarchy isn't selected as part of the table, the column is still included as part of the Copilot schema.
5. **Relationships always respected** - Relationships are still respected regardless of the AI data schema set. For example, if two fields are related, and one of the fields is included in the AI data schema, Copilot can still return answers that require that relationship.
6. **No end-user control** - End-users can't disable the AI data schema.
7. **Not a security feature** - Do not use AI data schema as a security feature, it's a suggestion to Copilot for the schema to use when answering questions

Verified answers

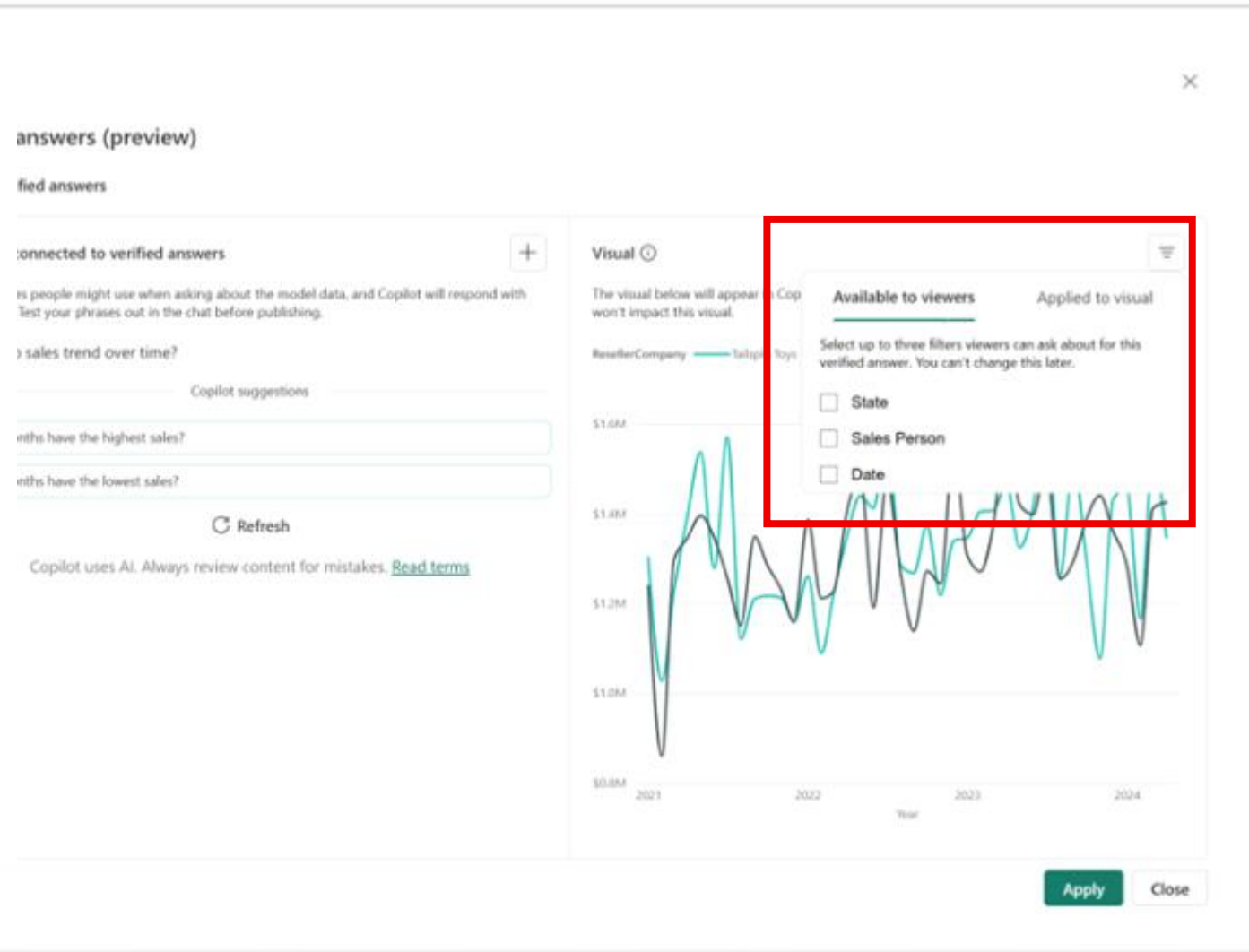
Ground Copilot and have Copilot provide answers defined by the model author

1. **Use both exact and semantic trigger phrases** – Provide 5–7 representative triggers per verified answer to cover different but valid ways users may ask the same question.
2. **Keep semantic matches realistic** – Synonyms, word order changes, and filters in prompts work, but adding/removing fields or swapping measures does not.
3. **Select supported visuals and models** – Verified answers only work with supported visual types (no AI visuals, decomposition tree, etc.) and supported models (Import, Direct Query, Composite, Direct Lake in service).

Verified answers (continued)

Ground Copilot and have Copilot provide answers defined by the model author

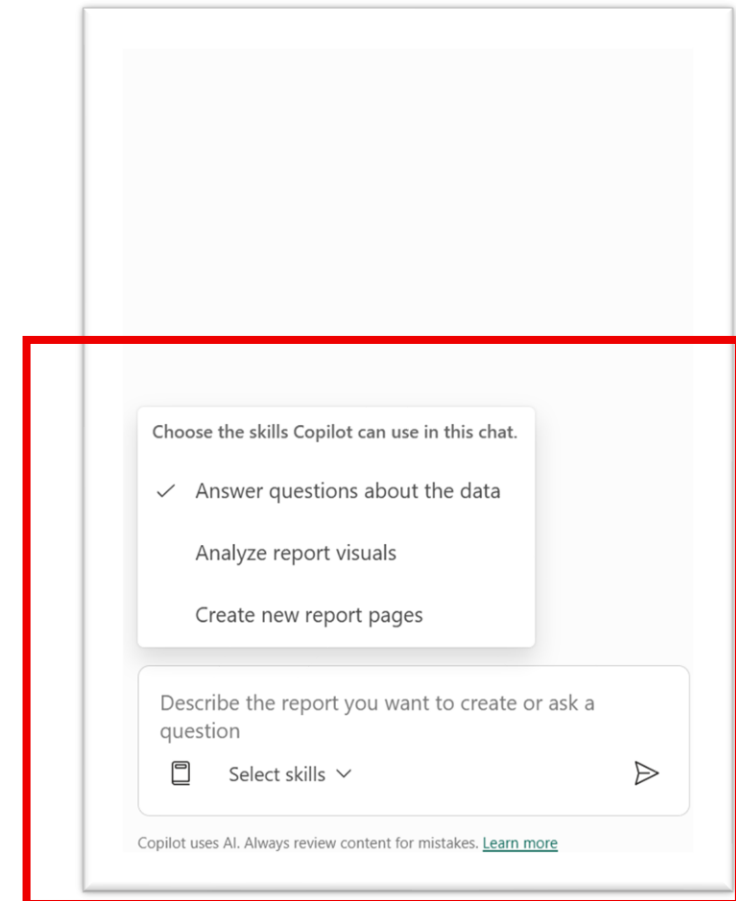
4. **Ensure fields and filters are visible** – Hidden fields or locked filters break verified answers. Make sure schema selection includes all fields you expect Copilot to use.
5. **Configure filters correctly** – Filters must be set to “All” in the report, visible to end users, and within the supported types (basic categorical, some advanced categorical, date range). Limit to 10 filters during creation, 3 at runtime



Verified answers (continued)

Ground Copilot and have Copilot provide answers defined by the model author

6. **Understand answer persistence** – Verified answers are tied to the semantic model, not the original report visual. Editing or deleting the report visual doesn't update the verified answer—manage updates in the Verified Answer setup dialog.
7. **Test carefully with the skill picker** – In desktop, disable authoring skills like “create page” to mimic end-user Copilot behavior and validate your verified answers trigger correctly.
8. **Communicate clearly with users** – Verified answers show a checkmark, matched phrase, and “how Copilot arrived at this.” Make sure these elements are accurate and trustworthy for confidence building.



Verified answers (continued)

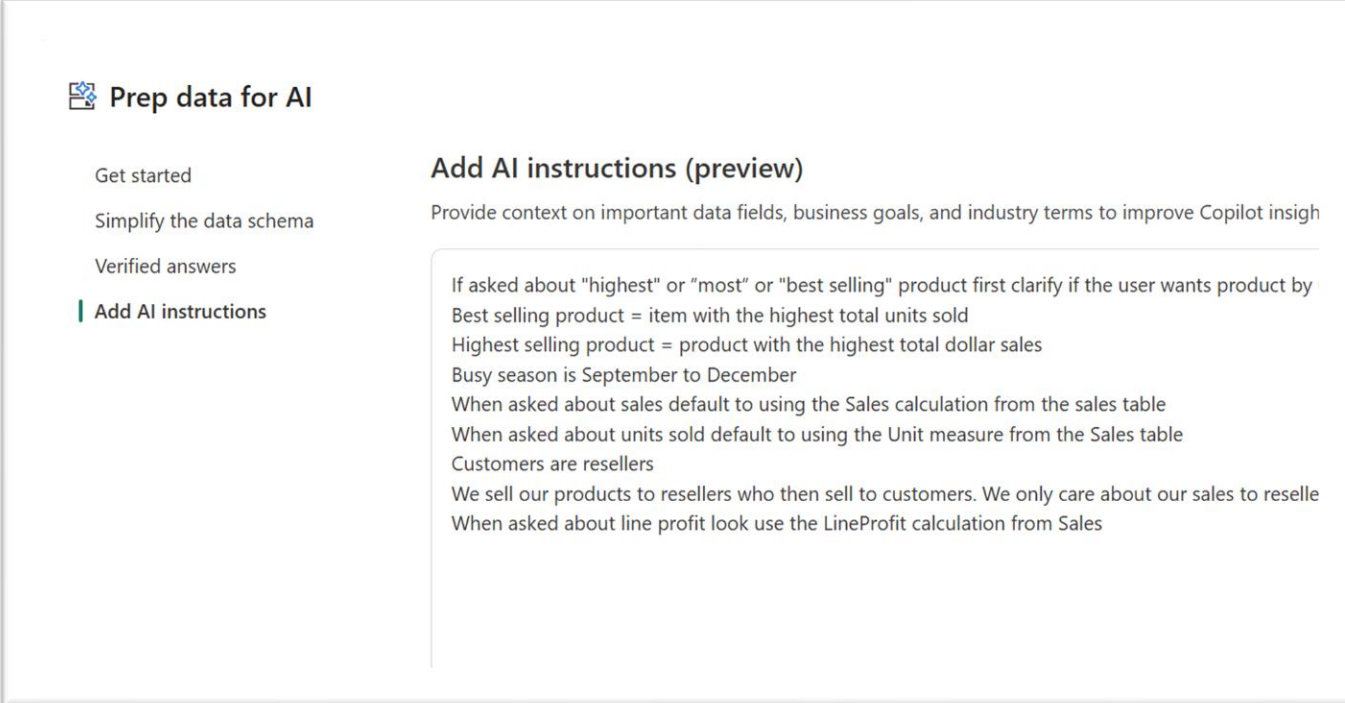
Ground Copilot and have Copilot provide answers defined by the model author

9. **Stay within limits** – Each model supports up to 250 verified answers, 15 trigger phrases per answer, and 500 characters per trigger. Respect these caps for reliability.
10. **Publish and validate in service** – After creating verified answers in desktop, publish to service and allow ~15 minutes for them to appear. A small edit in service (like tweaking a trigger phrase) can speed up sync.
11. **Select the visual first** – the “Set up Verified Answer” option only appears when a visual is selected.
12. **Manage through the Verified Answers dialog** – edits to report visuals won’t update Verified Answers; use the management dialog or recreate them

AI instructions

Give business context and instructions for Copilot to use to provide better answers

1. **Be explicit and specific** – Assume Copilot has no understanding of how to use the data model or business context for your data.
2. **Use analogies and descriptive language** to help the model understand your desired outcome.



 **Prep data for AI**

- Get started
- Simplify the data schema
- Verified answers
- Add AI instructions**

Add AI instructions (preview)

Provide context on important data fields, business goals, and industry terms to improve Copilot insight

If asked about "highest" or "most" or "best selling" product first clarify if the user wants product by
Best selling product = item with the highest total units sold
Highest selling product = product with the highest total dollar sales
Busy season is September to December
When asked about sales default to using the Sales calculation from the sales table
When asked about units sold default to using the Unit measure from the Sales table
Customers are resellers
We sell our products to resellers who then sell to customers. We only care about our sales to resellers
When asked about line profit look use the LineProfit calculation from Sales

AI instructions (continued)

Give business context and instructions for Copilot to use to provide better answers

- 3. Avoid ambiguity** – If something should be emphasized - or avoided - be clear. The more details and supporting content you can provide Copilot, the better.

Example: For Total Active Partners, use the measure "Monthly Active Partner Count" (do NOT filter on "Customers" table).

- 4. Group related instructions** – Organizing instructions by theme or purpose (that is, date logic, key metrics, industry terms) can help Copilot better understand your intent. Including structural elements, such as sections, hierarchies, and headers, can be very beneficial.
- 5. The order in which you author instructions can impact the output received** – Testing out different variations, orders, examples, and wording can help you get the correct outputs

AI instructions (continued)

Give business context and instructions for Copilot to use to provide better answers

6. **Break down complex instructions into simpler steps** to improve clarity and reduce errors.
7. **Keep instructions focused** – Depending on the model and instructions set, sometimes less is more. Conflicts and complexity in instructions can cause confusion for the large language model (LLM).

When I prepare my data for Copilot, which capabilities are affected?

Capability	AI data schemas	Verified answers	AI instructions	Descriptions
Get a summary of my report	No	No	Yes	No
Ask a question about the visuals on my report	No	Yes	Yes	No
Ask a question about my semantic model	Yes	Yes	Yes	No
Create a report page	No	No	Yes	No
Search	No	Yes	No	Yes
DAX query	No	No	Yes	Yes

FAQ and Troubleshooting guides for Preparing data for AI:
[Frequently Asked Questions about Preparing Data for AI - Power BI](#)

Please submit feedback



Are your semantic models ready for Copilot?